1. Constants & Relations (Knowledge Base Setup)

```
fruit(apple).
fruit(banana).
fruit(orange).

animal(dog).
animal(cat).
animal(cow).

color(blue).
color(green).
color(red).

/* output
?- fruit(X).
X = apple ;
X = banana ;
X = orange.

?- animal(dog).
true.

?- fruit(blue).
false.*/
```

2. Family Relationships

```
father(john,mary).
mother(mary,susan).
brother(peter,mary).
sister(susan,tom).

parent(X,Y):-father(X,Y);mother(X,Y).
grandparent(X,Y):-parent(X,Z),parent(Z,Y).
sibling(X, Y):-brother(X, Y);sister(X,Y);brother(Y,X);sister(Y,X).

/*output
?- grandparent(X, susan).
X = john ;
false.

?- sibling(peter, X).
X = mary ;
false.
*/
```

3. Likes and Preferences

```
likes(john,pizza).

likes(mary,pasta).

likes(tom,pizza).

likes(anna,pasta).

likes(anna,salad).


likes_pizza(X):-likes(X,pizza).


/* output

?- likes(X, pasta).

X = mary ;

X = anna.


?- likes(tom, salad).

false.


*/
```

4. Arithmetic & Operators

```
age(alice,20).
age(bob,17).
age(carol,22).

adult(X):-age(X,Age),Age>18.

/* output

?-adult(X).
X = alice ;
X = carol.

?-adult(bob).
false.

?- age(X,Age),Age<21.
X = alice,
Age = 20 ;
X = bob,
Age = 17 ;
false.
*/
```

5. Mini Problem (Knowledge Base + Rules)

```
male(john).
male(bob).
female(mary).
female(susan).
parent(john,bob).
parent(mary,bob).
parent(mary,susan).

father(X,Y):-parent(X,Y),male(X).
mother(X,Y):-parent(X,Y),female(X).
child(X,Y):-parent(Y,X).

/* output
?- mother(X, bob).
X = mary.

?- parent(X, susan).
X = mary.

?- female(X), child(X, _).
X = susan.
*/
```